DA3408 AI Operations — Module 1 Assignment

Experiment Management & Reproducibility with MLflow + DVC · Samrudh · `https://github.com/samrudh123/DA3408-Assignment-1`

Q1 — Technical Debt Diagnosis (written; full answer in `q1/aiops_q1.pdf`)

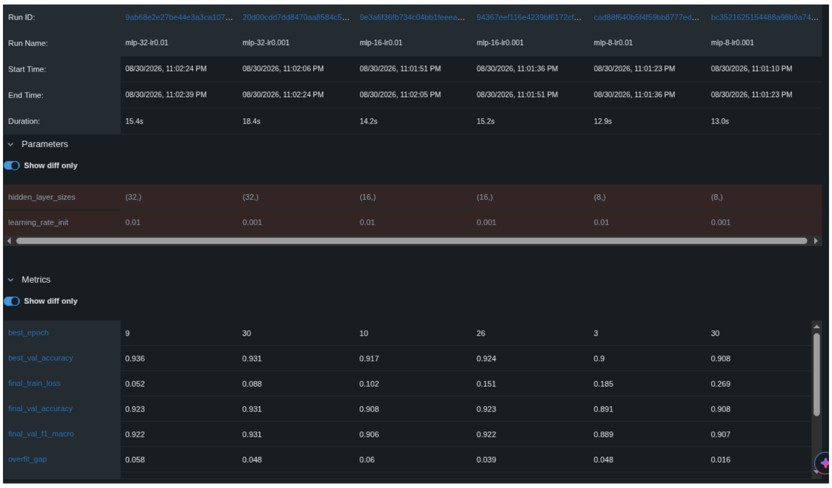
(a) Rounding change to *estimated delivery time* degrading *favourite restaurants* ⇒ **boundary erosion / entanglement**, the CACE principle: weights are learned jointly, so shifting one feature's distribution re-weights every other feature on retrain. The tell is that the effect was far larger than the code change.

(b) An unknown marketing team reading the model's output table ⇒ **undeclared consumers (visibility debt)**: with no declared interface the ML team cannot cost any schema or model change, and the campaign it drives risks closing a hidden feedback loop back into the training data.

(c) 14 undocumented shell scripts with no orchestration ⇒ **pipeline jungle**. *Mitigation:* replace them with a `dvc.yaml` DAG (ingest → clean → featurize → train → evaluate), rebuild with `dvc repro` so only changed stages re-run, version data via `dvc add` / `dvc push`, lift hyperparameters into `params.yaml`, and log every run to MLflow with a `git_commit` tag.

Q2 — MLflow Experiment Comparison (`q2/Q2_Mlflow.ipynb`)

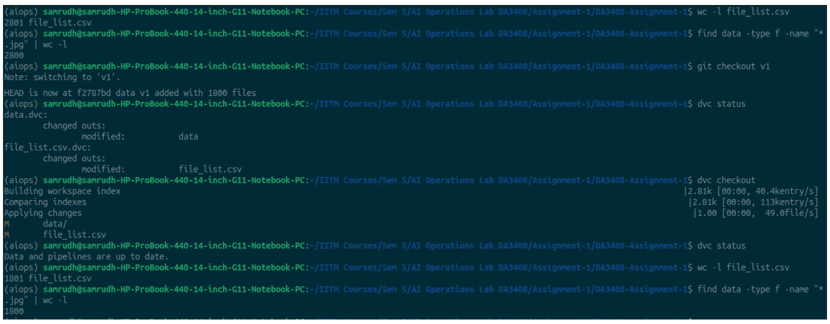
Six runs of an `MLPClassifier` on MNIST (12k subset, 30 epochs) over a 3 × 2 grid: `hidden_layer_sizes` $\in \{(8,),(16,),(32,)\}$ × `learning_rate_init` $\in \{0.001, 0.01\}$, every epoch logged with `mlflow.log_metric(..., step=epoch)` to a local server on port 4000.
Best run: `mlp-32-lr0.001` at **0.9313** val accuracy. Capacity drives the ranking — 32 beats 16 beats 8 at *both* learning rates.
Overfitting: `train_loss` and `val_accuracy` move apart. `mlp-32-lr0.01` reaches the lowest training loss of the sweep (0.052) yet validates *below* `mlp-32-lr0.001` (0.923 vs 0.931), whose loss is 1.7× higher. At 16 units, lr 0.01 cuts loss 0.151 → 0.102 and loses 1.5 points of val accuracy while the train/val gap widens 0.039 → 0.060.
Larger effect: architecture, by about 2× — varying width moves val accuracy 0.0271 on average, varying learning rate only 0.0138.



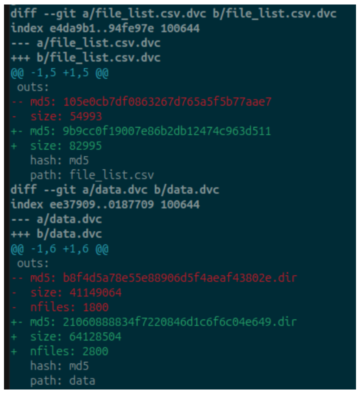
MLflow run-comparison table for all six runs, with Show diff only *enabled on both the parameter and metric panels.*

Q3 — DVC Data Versioning & Rollback (repo root; tags `v1`, `v2`)

`dvc init`, then `data.zip` → 1800 jpgs → `file_list.csv` (1801 lines = 1800 rows + header) → `dvc add data file_list.csv`, pushed to an **SSH remote**: `dvc remote add -d ssh-remote ssh://samrudh@10.17.64.87/research/.../dvcstore` followed by `dvc push -j 8` (key installed with `ssh-copy-id`; the keyfile lives in the `--local` config so no credential is committed). Tagged `v1`. Unzipping `new-labels.zip` gives 2800 jpgs and a 2801-line CSV; re-running `dvc add` rewrites only the md5 pointers, committed as "*data v2 with 2800 files*", tagged `v2` and `dvc push`ed again. **Rollback:** `git checkout v1` rewinds the pointers alone — `dvc status` then reports both outputs modified — and `dvc checkout` swaps the workspace bytes back, restoring **1801 lines / 1800 files** exactly.



Rollback proof: 2801 lines / 2800 files → `git checkout v1` → `dvc checkout` → *1801 lines / 1800 files.*



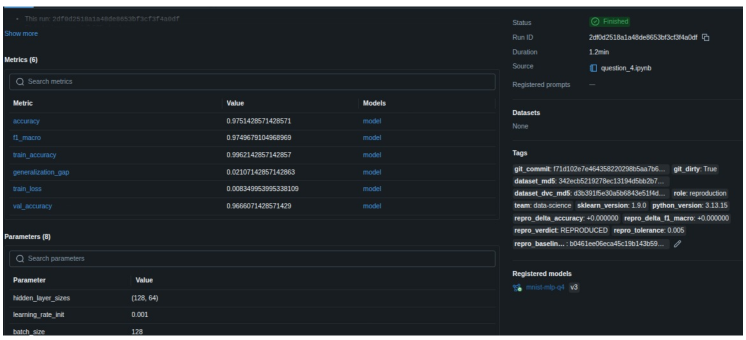
`git diff` of the `.dvc` pointers, v1→v2: `nfiles` 1800→2800.

Q4 — End-to-End Reproducibility Drill (`q4/question_4.ipynb`; recording `q4_video.mp4`)

Partner A trained `MLPClassifier((128,64))` on full MNIST with `seed=42`, logging params, metrics, per-epoch curves and provenance tags (`git_commit`, `git_dirty`, `dataset_md5`, `dataset_dvc_md5`, versions), versioned the data with DVC to an S3 remote, committed `data.dvc` *in the same commit* as the code, and registered `mnist-mlp-q4` into **Staging**. Baseline: `sujal_run.json`, tolerance ±0.005.
Partner B used only `git clone`, `git checkout`, `conda env create -f environment.yml`, `dvc pull` (restores `data/`) and a re-run of the notebook against `mlflow server --backend-store-uri sqlite:///mlflow.db`.

metric	baseline	repro	delta	ok
accuracy	0.9751	0.9751	+0.0000	yes
f1_macro	0.9750	0.9750	+0.0000	yes

Verdict: REPRODUCED. `dataset_md5`, `seed` and `sklearn 1.9.0` match; only `git_commit` and the Python patch release (3.13.14 vs 3.13.15) differ, neither of which moved the metrics.



Partner B's `reproduction` run: identical metrics, `repro_verdict`: REPRODUCED, `repro_delta_*`: +0.000000, registered as `mnist-mlp-q4` v3.